

Training

Configuration

The complete setting is stored in `configs/progressive_candidate_correction.yaml`.

Setting	Value
Random seed	45
Input size	384 x 384
Batch size	8
Epochs	120
Data-loader workers	2
Encoder	RepViT-M1.1, ImageNet pretrained
Optimizer	AdamW
Base learning rate	3×10^{-4}
Weight decay	1×10^{-4}
Adam betas	(0.9, 0.999)
Adam epsilon	10^{-8}
Mixed precision	enabled on CUDA
Gradient clipping	global norm 1.0
Validation	every epoch

```
python scripts/train.py \  
  --config configs/progressive_candidate_correction.yaml \  
  --data-root /path/to/dataset-root
```

Six-stage schedule

The optimizer and Adam moments are retained across stage boundaries. Trainable module groups and loss multipliers change together.

Epochs	Stage	Trainable modules
1-25	Part-structure information formation	shared decoder, Fish Region, endpoint orientation, and Part Map
26-50	Visual evidence formation	Redness, Shape, Lesion, Unaffected Surface, and candidate-path preparation

Epochs	Stage	Trainable modules
51-60	Part-route preparation	candidate correction, within-part context, Route assignment, and Spatial Support
61-85	Candidate and route interpretation	candidate correction, route-interpretation heads, and Route-wise composition
86-100	Semantic-output refinement	Route assignment, Spatial Support, route-interpretation heads, Route-wise composition, and final refinement
101-120	Joint fine-tuning	candidate correction, all route modules, and final refinement

The encoder is frozen throughout the six stages. Part-structure modules are trained in Stage 1. Visual evidence estimators are trained in Stage 2, then their tensors are detached from the candidate and route paths in Stages 3-6. The final stage jointly tunes the candidate, route, and refinement paths while preserving the established Part-structure and Visual evidence responses.

The exact objective multipliers are listed in [LOSS_FUNCTIONS.pdf](#).

Learning-rate schedule

Each stage starts with local linear warmup followed by cosine decay to 10% of the stage peak. The scheduler advances after every optimizer step and restarts at the next stage without resetting AdamW moments.

Epochs	Peak learning rate	Warmup
1-25	3.0×10^{-4}	2.0 epochs
26-50	3.0×10^{-4}	1.0 epoch
51-60	2.5×10^{-4}	1.0 epoch
61-85	2.0×10^{-4}	1.0 epoch
86-100	2.0×10^{-4}	1.0 epoch
101-120	7.5×10^{-5}	0.5 epoch

The encoder optimizer-group scale is 0.25 and the Route-wise composition scale is 0.75. During Stage 5, Route assignment, Spatial Support, route-interpretation heads, and Route-wise composition receive an additional 0.50 multiplier.

Augmentation

Geometric transforms are shared by RGB, semantic target, Fish Region, and Part Map. Color transforms affect RGB only.

Transform	Setting
Horizontal flip	probability 0.50
Shift	limit 0.03

Transform	Setting
Rotation	limit +/-15 degrees
Affine transform	probability 0.75
Color jitter	brightness 0.12, contrast 0.10, saturation 0.10, hue 0.03; probability 0.50
Brightness/contrast	limits 0.08/0.08; probability 0.30

Masks use nearest-neighbor interpolation. The dense targets in [SUPERVISION.pdf](#) are derived after augmentation.

Class weighting

Semantic and Part Map pixel frequencies are measured from the training masks. Inverse-square-root weights are mean-normalized and the background weight is multiplied by 0.25. The exact formula is given in [LOSS_FUNCTIONS.pdf](#).

Outputs and resume

Unless `--output-dir` is supplied, training writes to `outputs/progressive_candidate_correction/`:

- `resolved_config.yaml` contains the merged run configuration;
- `last.pt` is updated after every epoch; and
- `history.json` stores epoch, stage, training, and validation metrics.

Resume with the resolved configuration and checkpoint:

```
python scripts/train.py \
  --config outputs/progressive_candidate_correction/resolved_config.yaml \
  --data-root /path/to/dataset-root \
  --output-dir outputs/progressive_candidate_correction \
  --resume outputs/progressive_candidate_correction/last.pt
```

The checkpoint restores model, optimizer, stage-local scheduler, mixed-precision scaler, epoch, and metric history. The study endpoint is the epoch-120 `last.pt` checkpoint.

Smoke test

```
python scripts/train.py \
  --config configs/progressive_candidate_correction.yaml \
  --smoke-test \
  --device cpu
```

Candidate-correction comparisons use `configs/correction`. Semantic-segmentation comparisons use `configs/baselines`.